Jarn de Jong

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Quantum Physicist, Cryptographer, Developer

List of publications

Bio

Quantum physicist with a strong background in a wide range of topics. Software developer & tech enthusiast with a strong track record in Python, self-hosting, Docker and more.

Phd (summa cum laude) in quantum communication & cryptography, BSc. & MSc. in Applied Physics.

Programming & Computer skills

- Python
- Linux/bash
- Docker
- Networking
- Self-hosting
- Latex

Quantum skills

Networking & communication
Stabilizer & graph states
Multipartite entanglement
Local Equivalence
QKD & CKA
Anonymity in quantum networks
Photonic systems
Distributed systems
QECC & Fault-tolerance
State- and Process tomography
Quantum Internet use cases

Languages

Dutch	Native
English	C2
German	B1-B2

Experiences

2022-2025	Quantum Internet Alliance	Use Case team member
Project management, use case development and simulation, reporting.		
May 2023	LIP6 (Sorbonne Université)	Visiting researcher
Research visit at the LIP6 team. Present earlier work and collaborate with multiple researchers.		
2020-2025	Technische Universität Berlin	Scientific Employee
Designing & giving lectures, student supervision, scientific talks.		
2019-2020	QuTech	Research assistant & software developer
Hardware- and software development, simulations. Work span off into Q*Bird.		
2019	TNO	Quantum Technology Intern
Design and develop experiments for upcoming quantum device. Software development.		
2014-2015	Gymnasium Sorghvliet	Physics teacher intern
Half-year teaching internship for high school physics. Culminated in official qualification.		

Education and qualifications

2020-2024	TU Berlin, DE	PhD (Dr. Rer. Nat.)	*Summa cum laude*
Thesis under Anna Pappa in her quantum communication & cryptography group.			
2016-2019	TU Delft, NL	MSc. Applied Physics	*GPA: 8.4*
Quantum technologies track, thesis under Barbara Terhal at QuTech.			
2014-2015	TU Delft, NL	High school teaching qualification	*N/A*
High school physics teaching qualification with half-year internship at Gymnasium Sorghvliet during minor of BSc.			
2012-2015	TU Delft, NL	BSc. Applied Physics	*GPA: 7.5*
Thesis under prof. M. Verhaegen at Delft Centre for Systems and Control.			

Personal projects and extracurricular activities

	Software projects
Various software projects, e.g. the Graphstabilizer package and Spotify2PlexPlaylistSyncer.	
	Self-hosting
Deploying, hosting & orchestrating my website, multiuser Nextcloud-, Git-, media- and photo servers, and many more.	
2011-2025	Tutor, teaching (assistant), designing courses
High school tutoring, being teaching assistant, giving retake courses, designing and giving MOOC's.	
2012-2017	Various student committees
President of case tour- and symposium committee, member of various other committees.	

Links

Publications

Links	Year	Journal	Authors
Published / arXiv	2020	*PRX Quantum 1, 020325*	*F. Hahn, J. de Jong, A. Pappa*
			Anonymous Conference Key Agreement
Published / arXiv	2021	*New J. Phys. 23 083026*	*C. Thalacker, F. Hahn, J. de Jong, A. Pappa, S. Barz*
			Anonymous and secret communications in quantum networks.
Published / arXiv	2022	*PRX Quantum 3, 040306*	*F. Grasselli, G. Murta, J. de Jong, F. Hahn, D. Bruß, H. Kampermann, A. Pappa*
			Secure Anonymous Conferencing in Quantum Networks
Published / arXiv	2023	*Quantum 7, 1117*	*J. de Jong, F. Hahn, J. Eiser, N. Walk, A. Pappa*
			Anonymous conference key agreement in linear quantum networks
Published / arXiv	2023	*Phys. Rev. Research 5, 033222*	*L. Rückle, J. Budde, J. de Jong, F. Hahn, A. Pappa, S. Barz*
			Experimental ACKA using linear cluster states
Published / arXiv	2024	*Phys. Rev. Research 6, 013330*	*J. de Jong, F. Hahn, N. Tcholtchev, M. Hauswirth, A. Pappa*
			Extracting GHZ states from linear cluster states
arXiv	2024	(preprint) *arXiv:2410.03961*	*A. Burchardt, J. de Jong, L. Vandré*
			Algorithm to Verify Local Equivalence os Stabilizer States
Published / arXiv	2025	*Phys. Rev. A 111, 052449*	*L. Vandré, J. de Jong, F. Hahn, A. Burchardt, O. Gühne, A. Pappa*
			Distinguishing graph states by the properties of their marginals.
Published / arXiv	2025	*Phys. Rev. Applied 24, 054053*	*J. de Jong, S. Scheiner, N. Solomons, Z. Chaoui, D. Markham, A. Pappa*
			Anonymous and private parameter estimation in networks of quantum sensors

Selected talks

Links	Year	Type	Event	Location
Slides	2021	*PhD Seminar*	*Berlin School of Optical Sciences and Quantum Technologies*	Online
			Measurement-based and blind quantum computation	
Slides	2022	*Research visit*	*Koch group (Freie Universtät Berlin)*	Berlin, DE
			Graph states as quantum networks	
Slides	2023	*Conference*	*DPG spring meeting 2023, atomic physics & quantum information*	Hannover, DE
			Extracting maximal entanglement from linear cluster states	
Slides	2023	*Research visit*	*LIP6, Sorbonne Université*	Paris, FR
			Extracting GHZ states from linear cluster states - to extract or not to extract	
Slides	2023	*Meeting*	*Yearly Quantum Internet Alliance meeting*	Paris, FR
			Anonymity in quantum network protocols	
Slides / Gif	2024	*Invited speaker*	*MATH+ Einstein workshop on Quantum Cryptography and Quantum Networks*	Berlin, DE
			Quantum Networking using graph theory	